Computer Networking CoSc3211_

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Computer Networking

- Data: raw facts that are collected
- Information: processed data that enables us to take decisions

Computer Networking

- Data Communication: process of exchanging data or information
  - Characteristics
    - Delivery: data delivered to correct destination and user
    - Accuracy: deliver data accurately without any errors
    - Timeliness: audio and video delivered in time without delay (real time transmission)
    - Jitter: variation in packet arrival time

Computer Networking

- Data Communication: process of exchanging data or information
  - Components
    - Message: information to be communicated
    - Sender: any device that is capable of sending the data
    - Receiver: a device that the sender wants to communicate the data
    - Transmission Medium: path by which the message travels
    - Protocol: agreed upon set or rules used by the sender and receiver to communicate data

Computer Networking

- Data representation: text, numbers, images, audio, video
- Data flow:
  - Simplex (unidirectional): Only one of the devices sends the data and the other one only receives the data
  - Half Duplex: both the stations can transmit as well as receive but not at the same time.
  - Full duplex mode: both stations can transmit and receive at the same time.

Computer Networking

- Computer network: collection of an autonomous computing devices
  - Categories
    - Based on transmission technology
      - Packet switched
      - Circuit switched networks
    - Based on administration
      - Private
      - Public
    - Based on architecture
      - Client Server
      - Peer to Peer
    - Based on medium
      - Wired
      - Wireless
    - Based on size:
      - LAN
      - MAN
      - WAN

Computer Networking

- Protocol:
  - Defines:
    - what is to be communicated,
    - how it is to be communicated and
    - when it is to be communicated
  - Elements:
    - Syntax: structure or format of the data
    - Semantics: structure or format of the data, indicates the interpretation of each section
    - Timing: tells the sender about the readiness of the receiver

Computer Networking

- Standards: to ensure interconnectivity and interoperability between various networking hardware and software components or vendors
  - De facto Standard: have been traditionally used and mean by fact or by convention, not approved by any organized body but are adopted by widespread use.
  - De jure standard: by law or by regulation.

Computer Networking

- Standards:
  - Standard creation committees:
    - International Organization for Standardization(ISO)
    - International Telecommunications Union - Telecommunications Standard (ITU-T) American National Standards Institute (ANSI)
    - Institute of Electrical & Electronics Engineers (IEEE)
    - Electronic Industries Associates (EIA)

Computer Networking

- Standards:
  - Standard creation forums:
    - ATM Forum
    - MPLS Forum
    - Frame Relay Forum
  - Standard creation agencies:
    - Federal Communications Committee (FCC)

Computer Networking

- During transit data is in the form of electromagnetic signals
  - Analog data: information that is continuous
  - Digital data: information that has discrete states

Computer Networking

- Signals - Analog Signal: infinite values in a range.
  - many levels of intensity over a period of time
  - simple analog signal: is a sine wave that cannot be further decomposed into simpler signals characterized by three parameters:
    - Peak Amplitude:
      - amplitude: absolute value of its intensity at time t
        - proportional to the energy carried by the signal
      - peak amplitude: absolute value of the highest intensity

Computer Networking

- Signals – Analog Signal: infinite values in a range.
  - many levels of intensity over a period of time
  - simple analog signal:
    - Frequency: number of cycles completed by the wave in one second
      - Period: time taken by the wave to complete one second.
      - Wavelength: distance a signal travels in one period
        - Wavelength = Propagation Speed X Period
        - Wavelength =Propagation Speed X 1/Frequency
        - measured in micrometers
        - varies from one medium to another

Computer Networking

- Signals - Analog Signal: infinite values in a range.
  - many levels of intensity over a period of time
  - simple analog signal:
    - Phase: position of the waveform with respect to time
      - indicates the forward or backward shift of the waveform from the axis It is measured in degrees or radian

Computer Networking

- Signals – Analog Signal: infinite values in a range.
  - many levels of intensity over a period of time
  - simple analog signal
  - Composite Signal: combination of two or more simple sine waves with different frequency, phase and amplitude.

Computer Networking

- Signals - Digital Signal: limited number of defined
  - represented in the form of voltage levels
  - BIT LENGTH or Bit Interval (Tb): time required to send one bit.
    - measured in seconds.
  - BIT RATE: number of bits transmitted in one second.
    - expressed as bits per second (bps).
    - Bit rate = 1 / Bit interval
  - Baud Rate: rate of Signal Speed, the rate at which the signal changes.
    - A digital signal with two levels 0 and 1 will have the same baud rate and bit rate.

Computer Networking

- Periodic & Non Periodic Signals
  - <u>Periodic</u>: repeat itself after a fixed time period
  - <u>Non-Periodic</u>: do not repeat itself after a fixed time period

Computer Networking

- Transmission Channels:
  - <u>Low pass Channel</u>: has the lowest frequency as 0 and highest frequency as some non-zero frequency f1
  - <u>Band pass channel</u>: has the lowest frequency as some non-zero frequency f1 and highest frequency as some non-zero frequency f2

Computer Networking

- Transmission of Digital signal
  - Baseband Transmission: transmitted without making any change to it (i.e. Without modulation)
    - the bandwidth of the signal to be transmitted has to be less than the bandwidth of the channel
    - Bandwidth: difference between the highest and lowest frequency
    - Wideband channel: channel whose bandwidth is more than the bandwidth of the signal
    - Narrowband channel: channel whose bandwidth is less than the bandwidth of the signal

Computer Networking

- Transmission of Digital signal
  - Broadband Transmission: use modulation, i.e. we change the signal to analog signal before transmitting it
    - we have a bandpass channel we cannot directly send this signal through the available channel
    - the signal is modulated using a carrier frequency
    - signal is demodulated and again converted into an digital signal at the other end

Computer Networking

- Transmission impairments: transmission medium are not perfect, error is introduced in the transmitted data (received signal is not the same as the original signal)
  - Causes:
    - Attenuation: loss of energy due to distance
      - resistance of the medium ⇒ electrical energy in the signal may convert to heat.
      - amplifiers are used to amplify the signal.
    - Distortion: changes the shape of the signal.
    - Noise: unwanted signal that is mixed or combined with the original signal during transmission.

Computer Networking

- Transmission Media: anything that can carry information from a source to a destination
  - usually free space, metallic cable or fiber optic cable
  - Guided: cabling system that guides the data signals along a specific path
    - twisted-pair cable, coaxial cable transport signals in the form of electric signals
      - Twisted-pair cable: two conductors (normally copper), each with its own plastic insulation, twisted together.
        - Unshielded Twisted-Pair cable (UTP)
        - Shielded Twisted-Pair cable (STP)
      - Coaxial cable:- uses copper wire to conduct the signals electronically
    - fiber-optic cable transport signals in the form of light

Computer Networking

- Transmission Media: anything that can carry information from a source to a destination
  - Unguided: without using a physical conductor, uses wireless electromagnetic signals to send data
    - Radio waves: frequencies between 3 KHz and 1GHz, Omni-directional
    - Micro waves: between 1 and 300 GHz, can be narrowly focused, line-of-sight propagation
    - Infrared: 300 GHz to 400 GHz, short range communication, cannot penetrate walls

Computer Networking

- Transmission Media:
  - Propagation ways:
    - Ground-wave propagation
    - Sky-wave propagation
    - Line-of-sight propagation

Computer Networking

- Network Topology: description of the layout of the region or area covered by that network
- Connections
  - Point-to-point: provide a direct link between two devices
  - Multi-point connections: provide a link between three or more devices on a network.
    - Bus (point-to-multipoint) topology
    - Ring topology
    - Star topology
    - Hybrid topology
    - Mesh topology
    - Tree topology

Computer Networking

- Transmission Media:
  - Propagation ways:
    - Ground-wave propagation
    - Sky-wave propagation
    - Line-of-sight propagation

Computer Networking

- Network Devices
  - Repeater: operates only in the physical layer. A repeater receives a signal and, before it becomes too weak or corrupted, regenerates the original bit pattern.
    - does not actually connect two LANs; it connects two segments of the same LAN

Computer Networking

- Network Devices
  - Hub: used to connect segments (devices) of a LAN
    - hubs don't filter information;
    - hubs broadcast incoming packets to all computers connected to it
    - used in star or ring topology
    - broadcasting device

Computer Networking

- Network Devices
  - Hub:
    - Passive hub: just a connector.
      - It connects the wires coming from different branches.
      - In a star topology Ethernet LAN, a passive hub is just a point where the signals coming from different stations collide.
      - below the physical layer
    - Active hub: a multipart repeater.
      - connections between stations in a physical star topology.

Computer Networking

- Network Devices
  - Switch: switch filters and forwards data packets across a network.
    - keeps a record of the MAC addresses of the devices attached to it.
    - forwards the packet directly to the recipient device by looking up the MAC address.
    - intelligent and expensive than Hub
    - point to point communication device

Computer Networking

- Network Devices
  - Router: specialized network device, used to interconnect different types of network that uses different protocol.
    - allows the users to connect several LAN and WAN
    - use the routing table to determine how to forward packets.
    - router analyzes a destination IP address of a given packet and compares it with the routing table to decide the packet's next best path
    - shares information with other routers in networking.
    - Wireless router – offers Wi-Fi connectivity
    - Broadband routers / Broadband modem – provided by the internet service provider

Computer Networking

- Network Devices
  - Bridge: Connects two LANS having the same protocol
    - (e.g. Ethernet or Token ring).
    - filters content by reading the MAC addresses of source and destination.
    - does not filter broadcast
    - if data is not destined for other network, it is prevented from passing over the bridge
    - slower than repeater due to filtering

Computer Networking

- Network Devices
  - Gateway: connects networks that use different protocols.
    - Transport gateway: connects two computers that use different transport protocols, reformatting packets as need be
    - Application gateway: understands the format and content of the data and translates messages from one format to another,

Computer Networking

- Errors:
  - binary data:
    - Single bit errors: a bit value of 0 changes to bit value 1 or vice versa.
      - more likely to occur in parallel transmission.
    - Burst Errors: multiple bits of the binary value changes.
      - more likely to occur in serial transmission.
  - add some extra bits to the original data to detect and correct the errors
    - redundant bits which will be removed by the receiver
    - Detection methods:
      - Parity Check
      - Cyclic Redundancy Check
      - Checksum

Computer Networking

- Signal Encoding:
  - Digital Data to Digital Signal conversion:
    - Line Coding
    - Block Coding
    - Scrambling: not used

Computer Networking

- Analog data to analog signal conversion:
  - Modulation: over long distances
    - Analog modulation: analog audio data is modulated onto a carrier sine wave
      - Amplitude modulation (AM)
      - Frequency modulation (FM)
      - Phase modulation (PM).
    - Digital modulation: to convert digital data to analog signal
      - Amplitude Shift Keying(ASK)
      - Frequency Shift keying (FSK)
      - Phase Shift keying (PSK)

Computer Networking

- Analog to Digital Conversion using modulation:
  - Techniques:
    - Pulse Amplitude Modulation (PAM)
    - Pulse Code Modulation (PCM)
    - Pulse Width Modulation (PWM)

Computer Networking

- Network Models: operate networks, prominently the OSI reference model and the TCP/ IP Model
  - Open Systems Interconnection (OSI):
    - developed by International Organization for Standardization (ISO)
    - developed to allow systems with different platforms (Hardware & software) to communicate with each other

Computer Networking

- Network Models:
  - Open Systems Interconnection (OSI):
    - Layers:
      - Physical Layer: provides a standardized interface to physical transmission media
        - Representation of bits: transmission of signals from one device to another which involves converting data (1's & 0's)
        - Data rate: defines the data transmission rate
        - Synchronization of bits: sender and receiver have to maintain the same bit rate and also have synchronized clocks
        - Line configuration: point to point link, or a multi-point link
        - Physical Topology
        - Transmission mode: simplex, half duplex or full duplex mode.

Computer Networking

- Network Models:
  - Open Systems Interconnection (OSI):
    - Layers:
      - Data Link Layer: adds reliability to the physical layer by providing error detection and correction mechanisms:
        - Framing:
          - receives the data from Network Layer, divides the stream of bits into fixed size manageable units (Frames).
          - On the receiver side, the data link layer receives the stream of bits from the physical layer and regroups them into frames and sends them to the Network layer.
        - Physical Addressing: appends the physical address in the header of the frame

Computer Networking

- Network Models:
  - Open Systems Interconnection (OSI):
    - Layers:
      - Data Link Layer: adds reliability to the physical layer by providing <u>error detection and correction</u> mechanisms:
        - <u>Flow control</u>: sender sends the data at a speed at which the receiver can receive it
        - <u>Error Control</u>: imposes error control mechanism to identify lost or damaged frames, duplicate frames and then retransmit them.
        - <u>Access Control</u>: imposes access control mechanism to determine which device has right to send data in an multipoint connection setting.

Computer Networking

- Network Models:
  - Open Systems Interconnection (OSI):
    - Layers:
      - Network Layer: makes sure that the data is delivered to the receiver despite multiple intermediate devices
        - sending side:
          - accepts data from the transport layer,
          - divides it into packets,
          - adds addressing information in the header
        - uses logical address commonly known as IP address to recognize devices on the network
        - each packet is independent of the other and:
          - may travel using different routes to reach the receiver
          - may arrive out of turn at the receiver

Computer Networking

- Network Models:
  - Open Systems Interconnection (OSI):
    - Layers:
      - Transport Layer: process to process delivery of data and makes sure that it is intact and in order.
        - sending side:
          - receives data from the session layer,
          - divides it into units called segments
        - makes use of port address to identify the data from the sending and receiving process

Computer Networking

- Network Models:
  - Open Systems Interconnection (OSI):
    - Layers:
      - Transport Layer:
        - transported in a connection oriented or connectionless manner
        - responsible for segmentation and reassembly of the message into segments using sequence numbers
        - carries out flow control and error control functions; but unlike data link layer these are end to end rather than node to node

Computer Networking

- Network Models:
  - Open Systems Interconnection (OSI):
    - Layers:
      - Session Layer (network dialog controller): establishes a session between the communicating devices called dialog and synchronizes their interaction
        - sending side:
          - accepts data from the presentation layer
          - adds checkpoints to it called sync bits
        - receiving end
          - receives data from the transport layer
          - removes the checkpoints

Computer Networking

- Network Models:
  - Open Systems Interconnection (OSI):
    - Layers:
      - Application Layer: enables the user to communicate its data to the receiver by providing certain services

Computer Networking

- Network Models:
  - TCP/IP model:
    - currently being used on our systems
    - collection of protocols often called a protocol suite
    - existed even before the OSI model was developed

Computer Networking

- Network Models:
  - TCP/IP model:
    - Originally had four layers:
      - <u>Host to Network Layer</u>: combination of protocols at the physical and data link layers

Computer Networking

- Network Models:
  - TCP/IP model:
    - Originally had four layers:
      - Network Layer or IP:
        - network layer protocol and is responsible for source to destination transmission of data
        - connection-less & unreliable protocol
        - best effort delivery service: no error checking
        - cannot be considered weak and useless; since it provides only the functionality that is required for transmitting data thereby giving maximum efficiency

Computer Networking

- Network Models:
  - TCP/IP model:
    - Originally had four layers:
      - Network Layer or IP:
        - combination of four protocols:
          - ARP - Address Resolution Protocol: resolve the physical address of a device on a network, where its logical address is known
            - physical address: imprinted on the NIC or LAN card
            - logical address (IP address): the Internet Address used to uniquely & universally identify a device
          - RARP - Reverse Address Resolution Protocol: used by a device on the network to find its own Internet address when it knows its physical address

Computer Networking

- Network Models:
  - TCP/IP model:
    - Originally had four layers:
      - Network Layer or IP:
        - combination of four protocols:
          - ICMP - Internet Control Message Protocol: inform the sender about datagram problems that occur during transit
            - used by intermediate devices
          - IGMP - Internet Group Message Protocol: allows sending the same message to a group of recipients.

Computer Networking

- Network Models:
  - TCP/IP model:
    - Originally had four layers:
      - <u>Transport layer</u>: responsible for transmission of data running on a process of one machine to the correct process running on another machine

Computer Networking

- Network Models:
  - TCP/IP model:
    - Originally had four layers:
      - Transport layer:
        - transport layer contains three protocols:
          - TCP – Transmission Control Protocol: reliable connection-oriented
            - divides the data it receives from the upper layer into segments and tags a sequence number
            - receiving end reorders data

Computer Networking

- Network Models:
  - TCP/IP model:
    - Originally had four layers:
      - Transport layer:
        - transport layer contains three protocols:
          - UDP – User Datagram Protocol: simple protocol used for process to process transmission
            - unreliable, connectionless protocol for applications that do not require flow control or error control
            - adds port address, checksum and length information

Computer Networking

- Network Models:
  - TCP/IP model:
    - Originally had four layers:
      - Transport layer:
        - transport layer contains three protocols:
          - SCTP – Stream Control Transmission Protocol: relatively new protocol
            - combines the features of TCP and UDP
            - used in applications like voice over Internet and has a much broader range of applications

Computer Networking

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Computer Networking

- Network Models:
  - TCP/IP model:
    - Originally had four layers:
      - <u>Application Layer</u>: combination of Session, Presentation & Application Layers
        - defines high level protocols like:
          - File Transfer (FTP),
          - Electronic Mail (SMTP),
          - Virtual Terminal (TELNET),
          - Domain Name Service (DNS), etc.

Computer Networking

- IP Addressing:
  - Packets in the IPv4 format are called datagram
  - IP datagram consists of a header part and a text part (payload)
  - IPv4:
    - IP addresses: uniquely identifies a device on the Internet.
      - Every host and router on the Internet has an IP address,
      - encodes its network number and host number.
      - An IPv4 address is 32 bits long
      - used in the Source address and Destination address fields of IP packets.
      - An IP address does not refer to a host but it refers to a network interface

Computer Networking

- IP Addressing:
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Computer Networking

- IP Addressing:
  - IPv4:
    - Address Space: total number of addresses used by the protocol.
      - N bits to define an address ⇒ the address space is 2^N
      - 32-bit IPv4 addresses, which means that the address space is 4,294,967,296 (more than 4 billion).

Computer Networking

- IP Addressing:
  - IPv4:
    - Notations:
      - Binary notation: as 32 bits. ex. 11000001 10000011 00011011 11111111
      - Dotted decimal notation: easier to read, decimal point (dot) separating the bytes.
        - Each byte (octet) is 8 bits hence each number in dotted-decimal notation is a value ranging from 0 to 255.
        - Ex. 129.11.11.239

Computer Networking

- IP Addressing:
  - IPv4:
    - Classful addressing: the address space is divided into five classes: A, B, C, D, and E.
      - An IP address in class A, B, or C is divided into netid and hostid.
    - Subnetting: allows a network to be split into several parts for internal use but still act like a single network to the outside world.
      - router needs a subnet mask that indicates the split between network + subnet number and host.
      - Ex. 255.255.252.0/22
        - 22 to indicate that the subnet mask is 22 bits long.

Computer Networking

- IP Addressing:
  - IPv4:
    - CIDR: allocate the remaining IP addresses in variable-sized blocks, without regard to the classes.
    - NAT (Network Address Translation): scarcity of network addresses in IPv4 led to the development of IPv6. Due to this, we need to use private IP address inside the organization and translate it to public IP address using NAT.
    - IP Header: IPV4 has 32 bit header information such as Version, IHL, total length, types of services

Computer Networking

- Routing Protocols:
  - routing algorithm: determines the path for a packet
    - table types:
      - nonadaptive (static)
      - adaptive (dynamic)
    - protocol types:
      - Interior routing protocols
      - Exterior routing protocols
  - routing:
    - requires a host or a router to have a routing table which is constructed by the routing algorithm

Computer Networking

- Switching and Multiplexing:
    - switching: mechanism by which data/information sent from source towards destination which are not directly connected
        -

Computer Networking

- Switching and Multiplexing:
  - switching:
    - Circuit Switching: two communicating stations are connected by a dedicated communication path
      - consists of intermediate nodes in the network and the links that connect these nodes
      - steps:
        - Circuit Establishment: end-to-end connection before any transfer of data.
        - Data transfer: data may be analog or digital, depending on the nature of the network, generally full-duplex.
        - Circuit disconnect: Terminate connection at the end of data transfer.

Computer Networking

- Switching and Multiplexing:
  - switching:
    - Packet Switching: designed to address the shortcomings of circuit switching
      - communication is discrete in form of packets.
      - switching modes:
        - Cut-through
        - Store-and-forward
        - Fragment-free

Computer Networking

- Switching and Multiplexing:
  - Multiplexing
    - whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared
    - multiplexing: set of techniques that allows the simultaneous transmission of multiple signals across a single data link

Computer Networking

- Switching and Multiplexing:
  - Multiplexing
    - Frequency division Multiplexing (FDM)
      - analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted.
      - signals generated by each sending device modulate different carrier frequencies.
    - Wavelength-division multiplexing (WDM):
      - uses the high-data-rate capability of fiber-optic cable.
      - optical fiber data rate is higher than the data rate of metallic transmission cable.
    - Time division multiplexing
      - at different interleaved time intervals.
      - Each connection occupies a portion of time in the link.

Computer Networking

- Switching and Multiplexing:
  - Medium access control:
    - problem of controlling the access to the medium similar to the rules of speaking in a meeting,
      - two people do not speak at the same time;
      - do not interrupt each other;
      - do not monopolize the discussion;

Computer Networking

- Switching and Multiplexing:
  - Medium access control:
    - categories of multiple access protocols
      - Random Access Protocols – try your best like taxis do
        - MA – Multiple Access
        - CSMA – Carrier Sense MA
        - CSMA/CD – CSMA with Collision Detection
        - CSMA/CA – CSMA with Collision
      - Avoidance Controlled-Access Protocols – get permission
        - Reservation
        - Polling
        - Token Passing

Computer Networking

- Switching and Multiplexing:
  - Medium access control:
    - categories of multiple access protocols
      - Channelization Protocols - simultaneous use
        - FDMA - Frequency-Division MA
        - TDMA - Time-Division MA
        - CDMA - Code-Division MA